We Will Know: a reproducible pipeline for a code-animated, word- and beat-synchronised music video, with pre-registered validation of its timing components

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Abstract

We describe and evaluate an open pipeline that turns hand-written lyrics into a four-minute music video in which every frame is drawn in code.

Pipeline. One Suno V6 generation, obtained through the third-party sunoapi.org API, yields two sung takes and per-word timings. A rule fixed before generation selects one take. The written lyrics are sequence-aligned to the provider's word timings. Beats are tracked with librosa. A Remotion/Three.js composition then renders ten scenes, which illustrate ambitious interpretability and a scientific theory of deep learning. 86 figure cards are each timed to the sung word they illustrate and carry a provenance badge and a citation. Since v1.4.0 each card dives out of the scene object its lyric names, is shown at centre stage for at most 3.2 s, and docks back into that object, while the scenes draw concrete examples inside their general pictures. All visuals are pure functions of the frame number.

What we tested. Each component that decides *when* something happens on screen was tested against a criterion stated in advance.

1. On synthetic click tracks with known beats (12 tempos × 2 noise levels × 2 hop sizes × 5 seeds), the beat tracker at hop 256 placed beats within 15–19 ms (clean) of the truth. It failed the pre-registered criterion at 170 BPM through a half-tempo error. At librosa's default hop 512, the tempo estimates were quantized to errors of up to 3.55 BPM.
2. Against an independent recogniser (faster-whisper), the provider's word onsets agreed to a median of 0.235 s (take 1) and 0.172 s (take 2) over 393 and 401 matched words. Random ±1 s shifts gave 0.59 s and 0.50 s, so the agreement is not an artifact of the metric.
3. A plausible-looking timing-repair heuristic was **falsified** by the pre-registered test: on 43 words it moved onsets a median of 1.14 s *away* from the reference. A minimal post-hoc rule now touches only 2 grossly misplaced words.
4. A follow-up, also pre-registered, asked which system carries the Suno–Whisper lag. It separated the vocals (Demucs) and force-aligned the known lyrics on the vocal stem (MMS_FA). On the take the video then used, the provider's onsets were late by 0.137 s (95 % line-cluster CI 0.125–0.147 s, no drift over the song). Whisper was early by 0.080 s. The forced aligner was within 0.041 s of 34 blind hand-labelled onsets. The decision rule therefore shifted every word by −0.137 s, which cut the median error at the hand-labelled onsets from 0.233 s to 0.131 s (post hoc). The lag is take-specific: 0.061 s on the other take.
5. Every 1920×1080, 30 fps render matched the song's duration to within 0.03 s (7213 frames for the original take, 7212 for the edited one).
6. To take one word ("mind") out of the lyrics, one line was re-sung with the provider's replace-section operation. The line was fixed: an independent recogniser now hears the new word, and coverage stayed at 430/430. A pre-registered check that the rest of the take was untouched **failed**. The provider regenerated about 30 s around an 11.6 s window, shifted the whole file by 17 ms, and returned word times copied from the parent take. Re-running exp05's rule on the edited take gave a lag of 0.156 s [0.144, 0.167], which the video now corrects.
7. The figure cards reuse data released by the original authors or well-known third parties: 31 pinned files, with no experiment re-run. Two pre-specified consistency checks passed. The five largest Fourier frequencies of the released grokking embedding equal the paper's key frequencies, and ARENA's top induc-

tion heads include the IOI paper's. Every sung line has a card (56/56). Paired frames carry more detail than v1.2.0: median $\log_2$ PNG-size ratio +0.26, section-bootstrap 95 % CI [+0.18, +0.42]. A fact-check removed three overclaims from our own plan.

8. Staging the cards inside the scenes (v1.4.0) passed its pre-registered non-inferiority test against v1.3.0. The median $\log_2$ PNG-size ratio was −0.014, with a section-bootstrap 95 % CI of [−0.075, +0.016] against a margin of −0.10. The change was not uniform: verse 1 lost detail (−0.19) and the finale gained (+0.13). Partly, this may be the dimming of the scene behind the hero, which was not tested. A figure is the hero during 82.6 % of the sung time, covers at most 32.8 % of the frame, and 71 of the 86 cards dock into a scene object.
9. In v1.5.0 each scene owns its figures: at most two panels in slots reserved away from its key objects, and ideas the scene can draw itself drawn in place; the outro became a flying rewind through the whole video. Pre-registered layout checks passed: every panel stays ≥ 3 s (one cut-limited panel 2.76 s), at most two at once, and no slot overlaps a protected object. The median time a figure is on screen rose from 2.2 s as v1.4.0's hero to 6.0 s. Detail before the outro was not lower than v1.4.0 (+0.027 [−0.049, +0.088]). The pre-registered check that the 0:14 window is as full as v1.3.0 **failed** on the first render (−0.58 $\log_2$). A post-hoc fix (verse 1's layers flying out behind the tokens card, part of the approved plan but skipped) brought the window to −0.09 in the released render. That is much fuller than v1.4.0 (+0.26 to +0.83) but still short of the criterion, which was not changed. Final detail before the outro: +0.033 [−0.047, +0.102].

We report these as properties of this pipeline and this song. They are not general claims about the music model.

1. Introduction

Context. In 2026 a genre of short videos appeared in which a language model writes the code for every frame of an animated music video, usually with a song from Suno and React-based rendering through Remotion (e.g. the prompt collection *awesome-opus5-5-videos* on GitHub). Such videos stand or fall on *timing*: lyrics must light up when they are sung, and cuts must land on beats.

The question. Public descriptions of these videos report which tools were used. They do not report whether the timing is right, or how one would know.

The subject of the song. The song, *We Will Know*, is an anthem for an ambitious science of information processing. Its themes:

- the goal of fully reverse-engineering neural networks (leogao, 2025; Goodfire, n.d.);
- the *ladder of abstraction* from neurons and attention heads to behaviors and drives (Goodfire, n.d.);
- the five directions toward a scientific theory of deep learning (Simon et al., 2026);
- an atlas of the mathematics involved;
- "ideal solving": the shortest theory that predicts any behavior and any intervention (Kolmogorov, 1965; Pearl, 2009).

Contributions.

1. An open, single-command, reproducible pipeline: song, word timings, beats, timeline, deterministic render, automated verification. Paid generation is cached and tests run offline.
2. Pre-registered checks of the three timing components (take selection, beat tracking, word timing), each with an explicit falsification criterion, raw data and a control.
3. Two negative results reported in full:
   - a beat-tracker half-tempo failure inside the tested tempo range;
   - a word-timing repair rule that the test showed to be harmful.

   The repair finding changed what ships: without the test, 27 words of the video would have lit up 0.8–2.5 s late.
4. A pre-registered decomposition of a timing disagreement between two black-box systems. A third, lyrics-aware reference and blind hand-labelled anchors attribute it to both: the provider is late by 0.137 s and Whisper is early by 0.080 s. The resulting correction was applied to the video.
5. A pre-registered check of a one-line lyric edit made through the provider's section replacement. It shows that the edit reaches well beyond the requested window and invalidates per-take timing calibrations,

which are re-estimated rather than assumed.
6. An illustration track whose "released data" cards are derived from pinned, hashed artifacts published with the cited papers, rather than from re-run experiments or invented plots. Each card carries a provenance badge, and the derived numbers were checked against the papers before release.

2. Background and related work

- **Beat tracking.** librosa (McFee et al., 2015) implements dynamic-programming beat tracking (Ellis, 2007). It builds an onset-strength envelope, picks a global tempo from an autocorrelation tempogram with a log-normal prior centred at 120 BPM, then places beats by dynamic programming. Beat accuracy is conventionally scored with an F-measure at a ±70 ms tolerance, the default in `mir_eval` (Raffel et al., 2014). Tempo estimation suffers from *metrical-level* (octave) ambiguity: half or double tempo is a common failure mode.
- **Lyric/word timing.** Whisper (Radford et al., 2023) yields word timestamps through cross-attention alignment. These are known to be imprecise, which motivated forced-alignment add-ons such as WhisperX (Bain et al., 2023). We therefore use Whisper only as an *independent, noisy* reference, never as ground truth.
- **Forced alignment and separation.** A forced aligner is given the transcript and only has to find *when* each word occurs. Here we use the CTC aligner of Massively Multilingual Speech (MMS_FA; Pratap et al., 2023), as packaged in torchaudio. CTC posteriors are "peaky", which can shift boundaries by a few frames (Huang et al., 2024).
  - Lyrics-to-audio alignment on polyphonic music works better on a separated vocal stem. Demucs-based systems reach mean word-onset errors of about 0.3 s on JamendoLyrics (Demirel et al., 2021), and the best systems reach below 0.2 s (Durand et al., 2023).
  - We separate with Hybrid Transformer Demucs (Rouard et al., 2022).
- **Programmatic video.** Remotion renders React components frame by frame in headless Chrome. A video is a pure function of the frame index, which allows parallel, deterministic rendering. Three.js scenes run via `@remotion/three` with the ANGLE GL backend.
- **Scientific content shown in the video.** Flat minima and generalization (Hochreiter & Schmidhuber, 1997; Keskar et al., 2017); circular features for days and months (Engels et al., 2024); linear space and time representations that are more accurate in larger models (Gurnee & Tegmark, 2023, §3.1); function vectors (Todd et al., 2023); the five research directions of "learning mechanics" (Simon et al., 2026). Since v1.3.0, each idea is shown on a figure card. The card uses released data where the authors or a well-known third party published it (§3.6), and is otherwise drawn as a labelled schematic or computed live from the formula.

3. Method

3.1 Pipeline

A Python wrapper (`uv run anthem`) runs four phases. The phases communicate only through files.

1. **song.**
   - Sends one `POST /api/v1/generate` with custom mode, V6, the hand-written lyrics (2701 characters, 11 sections, 58 lines, 430 words), a style string, negative tags and `duration: 240`.
   - Polls `record-info` until `SUCCESS`.
   - Downloads both takes and covers, and calls `get-timestamped-lyrics` for each take.
   - Caches everything, so no second paid call is ever made implicitly. All requests and responses are logged in full.
2. **timeline.**
   - Parses the lyrics, selects a take (§3.2), aligns the words (§3.3) and tracks beats (§3.4).
   - Starts each section at its first sung word minus 0.1 s, snapped to a beat if one is within ±0.3 s.

- Writes `timeline.json`: sections and scenes, lines, words, beats, downbeats, and per-frame RMS and onset strength.

3. **render.**
   - Remotion 4.0.536 reads the timeline in `calculateMetadata`, so the video length equals the song length.
   - It stacks a starfield, faint drifting card titles, a spectrum ring, one scene per section, the figure cards (§3.6), a karaoke overlay (or giant kinetic words in the choruses), beat and cut flashes, a HUD and a citation ticker.
   - The scene layer punch-zooms on downbeats and is RGB-split for 0.25 s after each cut.
   - Render settings: `--gl=angle`, concurrency 8.
4. **verify.**
   - ffprobe acceptance checks, plus one still per section extracted with ffmpeg.

3.2 Take selection (fixed before generation)

The chosen take maximizes *coverage*, then minimizes `hootCer` (the provider's alignment score; lower is assumed better).

- **Coverage** is the share of written words, after normalization (lowercase, alphanumerics only), matched exactly to a provider word with `success=true`.
- **Why fix it.** Fixing the rule removes the freedom to pick whichever take makes the video look best.

3.3 Word alignment and repair

- **Cleaning and alignment.** Provider tokens arrive with section tags and line breaks glued on (e.g. `"[Verse 1]\nA "`). We strip the tags, split on whitespace, divide each entry's span evenly among its words, and align the result to the written words with `difflib.SequenceMatcher`.
- **Unmatched words.** Equal-length replace blocks are timed but not counted as matches. Deleted words are interpolated between their neighbours.
- **Repair v1** (pre-registered in exp03). A mid-line word longer than 1.2 s had its start moved to 0.5 s before its end. Any mid-line gap over 0.6 s pulled the earlier word up to its successor.
- **Repair v2** (post-hoc, §4.3). Only mid-line words longer than 3.0 s are moved. A preceding word follows only if its successor was moved.

3.4 Beats

- **Tracking.** `librosa.beat.beat_track` on the onset-strength envelope at 22.05 kHz with hop 256, chosen after the first unit test failed at hop 512 (§4.2).
- **Tempo.** Reported as 60 / median inter-beat interval.
- **Downbeats.** Every 4th beat, phase chosen by the summed onset strength. This is a heuristic that assumes 4/4.

3.5 Parameters and their rationale

Parameter	Value	Rationale
Song duration	240 s	~4-minute anthem; the API default of 20 s must be overridden
Video	1920×1080, 30 fps	YouTube 1080p; 33.3 ms frame = the timing resolution the viewer sees
Hop size	256 samples (11.6 ms)	Exp02: beats within 1 video frame; half the tempo-quantization error of 512

Parameter	Value	Rationale
Section lead / snap window	0.1 s / ±0.3 s	Cut slightly before the first sung word; never move a cut more than ~half a beat
Repair v2 thresholds	3.0 s; follow if gap > 0.6 s; repaired duration 0.5 s	3.0 s is more than 6 beats at 129 BPM; 0.5 s is about one beat
H1 threshold	0.25 s	About half a beat; set before running, given the expected noise in Whisper's timestamps
Word-time offset (`VIDEO_WORD_OFFSET_S`)	−0.156 s (code default 0)	Exp05's pre-registered decision rule, re-run on the edited take in exp06 (−0.137 s on its parent, take 1)
Replace-section window (exp06)	25.97–37.60 s (lines 4–6)	Gap midpoints around the re-sung lines; the API needs ≥ 10 s, and lines 5–6 alone span 7.6 s
MMS_FA windows (exp05)	30 s + 2 s context per side	Avoids full self-attention over 12 000 frames; the same scheme as `ctc-forced-aligner`
Exp05 decision thresholds	$\|\delta\| \geq 1$ frame (33 ms); $\|\beta\| \leq 0.05$ s; ≥ 10 anchors	One frame is the smallest error the viewer can see; 1.5 frames of bias is tolerated in the reference
Beat F window	±70 ms	`mir_eval` default
Seeds (exp02)	5 per cell	Enough to expose noise sensitivity; the observed between-seed SD was about 0

3.6 Illustration track and released data (v1.3.0)

Placement. - `content/vignettes.toml` declares 86 cards. Each has a trigger regex on a sung word (optionally its n-th match in the section), a lane, a title, text, a provenance badge and a citation. In v1.3.0 the lane was a named screen box (a slot). Since v1.4.0 it is only a concurrency cap: four lanes, so at most four cards are active at once (§3.7). - A card appears 0.15 s before its word. - It ends at its section's cut, unless that would leave it on screen for less than 2 s. - Of two cards in the same lane, the earlier ends where the later begins. - Placement fails loudly if a trigger is never sung. - Coverage is the share of sung non-backing lines with a card triggered by one of their words. Its pre-specified target was 1.0.

Badges. - *released data*: read from files released by the authors or a well-known third party. - *published numbers*: quoted from a paper's text or tables. - *live math*: computed in the video. - *from the paper*: a formula or claim. - *schematic*: our illustration.

Data rule (set by the author). - Never re-run an experiment whose models, results or data exist. - Prefer the original authors and well-known third parties. - Retrain only if nothing usable exists. Nothing had to be retrained.

`scripts/vignette_data.py` downloads 31 pinned files (523.8 MB) and records URL, revision, licence, size and sha256 for each. Downloaded `.pth` files load with `weights_only=True`. A pickle loads only through an unpickler that refuses every global except numpy reconstruction.

The derived card data comes from these sources: - the Nanda et al. (2023) mainline grokking run: loss curves, and the Fourier spectrum of the embedding every 1,000 epochs; - the three attribution graphs behind Lindsey et al. (2025), collapsed to the authors' supernodes (also used to drive the chorus circuits); - ARENA's per-head induction scores for GPT-2 small; - Neuronpedia SAE latents for the days of the week (Engels et al., 2024); - Epoch AI's model dataset; - Epoch's 245 points extracted from Hoffmann et al. (2022), drawn with both published parametric fits (Hoffmann et al.; Besiroglu et al., 2024); - Nakkiran et al.'s (2019) double-descent runs; - Gurnee

& Tegmark's place coordinates; - Todd et al.'s antonym pairs; - Arditi et al.'s refusal scores; - WeightWatcher's per-layer exponents (Martin & Mahoney, 2021); - the saved toy-superposition figure in Elhage et al.'s (2022) notebook; - Goodfire's block-sparse-featurizer figure; - the GPT-2 tokenizer.

3.7 Hero shots and concrete layers (v1.4.0)

In v1.3.0 the cards were boxes in the corners of the frame, beside the animation. In v1.4.0 they are staged inside the scenes. Each card's state is a pure function of song time t (`video/src/stage/shots.ts`), built from eased steps (smoothstep, 0.4 s):

- **emerge**: rises from 0 to 1 over 0.4 s after the card's start;
- **rank**: rises by 1 when the card's 3.2 s of hero time run out, and by 1 for each newer card of the same section that is active;
- **dock**: rises from 0 to 1 over 0.5 s after the card's end.

A card's *heroness* is emerge × (1 − min(1, rank)) × (1 − dock), and the card with the largest heroness is the hero. The drawn pose blends between four targets:

- the **anchor**: the scene object the lyric names, taken from geometry the scene exports, such as the rung sung last, the theory pillar of the sung direction, or the rotating atlas star of the sung field;
- the **hero pose**: centre stage at scale 1.8, or 1.38 in the two scenes whose giant lyrics take the lower third;
- an **orbit pose** per rank, smaller, tilted and blurred;
- the **dock**: a thumbnail at the anchor, or, in the three fly-through scenes, past the camera.

While a hero is up, the scene layer is scaled by at most 1.08 about the hero's anchor and dimmed to 72 % brightness. The intent is that the general picture stays visible around the example rather than being replaced by it.

The scenes also gained concrete layers inside their general pictures: - **Layer tunnel.** GPT-2 token ids of the sung line, from the pinned tokenizer. - **Loss surface.** The surface's own saddle and minima, found by Newton's method. The saddle's Hessian eigenvalues are −0.27 and +0.19. The sharp and flat basins have λmax 1.76 and 0.27, and both are shaken by ρ = 0.8, SAM's sharpness radius (Foret et al., 2021). The released grokking and double-descent curves are drawn as ribbons. - **Chorus.** The attribution graphs' real prompt tokens, ending in the model's output. - **Manifolds.** Month and year labels, the superposition pentagon, Todd et al.'s antonym pairs on the function vector, a true/false split (Marks & Tegmark, 2023), and the published Llama-2 world-probe R^2 per model size. - **Ladder.** An encyclopedia that files the rung figures. - **Finale.** The 280-bit string collapsing into its program, and a frame-wide do()-graph. - **Outro.** Every figure title spiralling into the token.

Labels that a hero would cover are drawn as margin callouts with leader lines.

3.8 Figures as part of their scenes, and the rewind (v1.5.0)

After v1.4.0 the author reported that the figures still read as "two overlapped independent videos". The causes were measured from the timeline: - **Short hero times.** 41 of 86 cards were the hero for under 2 s, because the newest card always took the centre. - **Covered key objects.** The hero covered a key scene object the whole time it was up: the ladder, the theory blocks and the closing statement. - **Empty stretches.** There were empty stretches, for example 14–18 s.

v1.5.0 replaces the hero model with a layout contract (`video/src/stage/layout.json`). Each scene declares: - at most two panel slots; - the protected rectangles of its key objects; - its lyric band.

Slots are tested to lie on screen, to be readable (≥ 0.9 of the native 600 × 350 card) and to be disjoint from the protected regions and bands. Placement then works as follows: - **Panels** (lanes a/b) stay in their slot for their whole time and are wired by a connector to the object they explain. The objects are geometry the scene exports: rung label ends, star projections, surface points, manifold centres. - **Lane `scene`** marks figures the scene draws itself, in place. - **A `[[clear]]` cue** ends a section's figures before a highlight line. - **The curation** (86 → 82 cards, logged with reasons in exp09) gives every panel ≥ 3 s while keeping a figure on every sung line.

The rewind. The outro is a pure schedule (`stage/rewind.ts`). Cuts sit on the beat grid and shrink from 1 beat to ¼ beat. Each cut plays an earlier moment of the video backwards at 2× rising to 12×. The moments are each figure's midpoint and each section's, latest first. The landing cut plays the intro backwards onto its single token. The copies are Remotion `<Freeze>` instances of the same `World` component that draws the live video (Remotion, 2026), with at most two on screen at once.

4. Experiments and results

4.1 Generation and selection (exp01)

- **Generation.** One task took 62 s from submission to `SUCCESS` and returned two takes: 240.40 s and 239.92 s.
- **Alignment entries.** The provider returned 431 and 439 aligned entries, with `success=true` for all of them.
- **Selection.** After cleaning, both takes covered 430/430 written words, so coverage tied at 1.0. The tie-break chose take 1 (hootCer 0.155 vs 0.165). Coverage turned out to be uninformative for this provider, and we do not claim that take 1 is musically better.
- **Section order.** The section boundaries recovered from take 1 follow the written section order.
- **Artifacts.**
  - Take 2 aligned its first line twice.
  - In take 1, "matrix" spanned 7.74 s and began inside the intro.

4.2 Beat tracker vs ground truth (exp02)

Discovery. A unit test written before the implementation (a 120 BPM click track, ±2 BPM) failed with 117.45 BPM. At hop 512 the tempogram's integer lag is 22 frames, and $43.07 \cdot 60 / 22 = 117.45$.

Systematic run. 30-s click tracks at 70–180 BPM. Pre-registered criterion: at hop 256 on clean clicks, every run must have $|\text{error}| \le 2$ BPM and $F \ge 0.95$.

- **Overall: FAIL.** 11 of 12 tempos passed.
- **The failure is a half-tempo lock.** At 170 BPM the tracker reported 85 BPM, with $F = 0.661$ (precision 1, recall 0.5). The same failure occurred at 180 BPM at hop 512 and on noisy clicks at hop 256.
- **Hop 512 quantization.** Tempo errors reached ±3.55 BPM.
- **The estimator does not remove it.** The median-spacing and tempogram estimators were identical in every cell, both locked to whole-frame periods. This corrects a claim in our own code comments.
- **Post-hoc slope estimator.** A least-squares slope of beat time vs index reduced the error to ≤ 0.01 BPM wherever the metrical level was correct. Since it was found post hoc, the pipeline keeps the pre-specified estimator.
- **Timing precision.** At hop 256, median beat deviation was 15–19 ms (clean) and 23–25 ms (noisy), below one frame.
- **Seed variance.** It was about 0 for tempo and at most ±0.021 for F.
- **The song.** It measures 129.3 BPM (506 beats), inside the validated range. Suno's own style tag says 124 BPM, which argues against a half or double error.

4.3 Word-onset agreement and the repair (exp03)

Pre-registered design.

- faster-whisper `small.en` (CPU int8) transcribed each take's full mix. It has no access to the lyrics.
- Written and recognized words were matched by text (exact difflib blocks).
- **H1:** median $|\text{onset} - \text{Whisper onset}| \le 0.25$ s per take.
- **H2:** on repaired words, the repair reduces $|\Delta|$ (one-sided Wilcoxon, $\alpha = 0.05$).
- **Control:** $U(-1, 1)$ s random shifts.

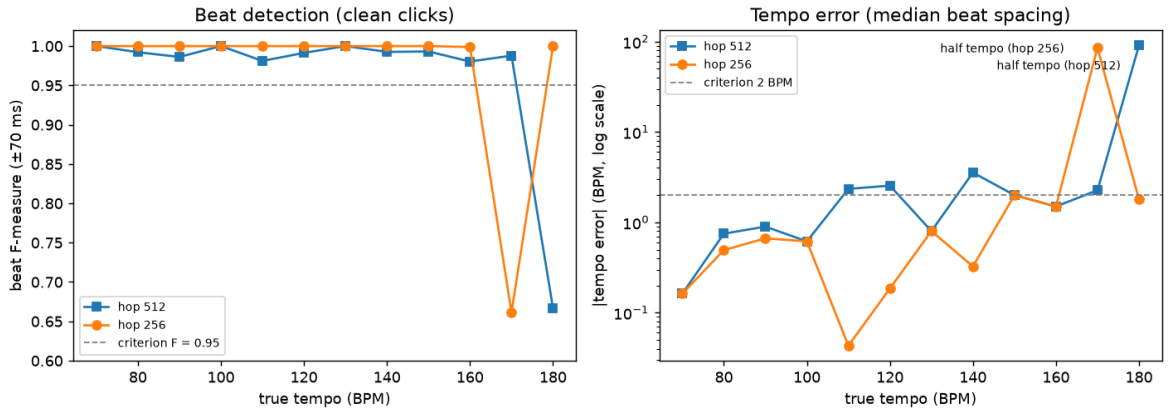


Figure 1: Exp02. Left: beat F-measure; right: absolute tempo error on a log scale, for clean click tracks. Half-tempo failures are labelled.

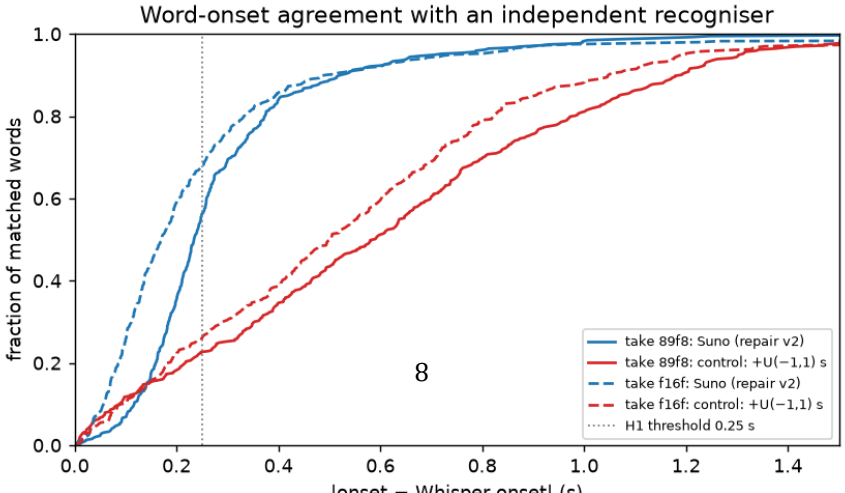
	Take 1 (video)	Take 2
Whisper words / matched pairs	440 / 393	435 / 401
Median |Δ| raw [95% bootstrap CI]	0.235 s [0.225, 0.248]	0.172 s [0.153, 0.190]
Median |Δ| repair v1	0.242 s [0.227, 0.257]	0.179 s [0.158, 0.195]
Median |Δ| repair v2	0.235 s [0.225, 0.248]	0.172 s (no repairs)
Median |Δ| random-shift control (with v2)	0.590 s	0.500 s
Share ≤ 0.25 s / ≤ 0.5 s (v2)	56.2 % / 88.8 %	67.8 % / 90.0 %

H1 passed for both takes.

H2 was falsified for repair v1. Over 43 repaired and matched words (both takes), the median paired change was −1.14 s, i.e. *worse*, with Wilcoxon p ≈ 1.0.

- **Diagnosis.** The 1.2 s threshold flagged real sung notes, e.g. "go" at 1.64 s. Shortening them opened gaps that cascaded the earlier words of those lines 0.8–2.5 s late (27 words in take 1). Whisper and the raw timings agreed on the true positions.
- **v2.** It touches only "A" and "matrix" in take 1, and moves both toward the reference: |Δ| 6.33 → 1.07 s and 6.81 → 0.43 s.
- **v2 is post-hoc and descriptive only.** The intended held-out take (take 2) contains no word that v2 would repair, so no confirmatory test was possible.

Post-hoc analysis. The signed differences show the disagreement is mostly a constant lag: Suno is later than Whisper for 98.5 % (take 1) and 93.8 % (take 2) of words, by a median of 0.234 s and 0.165 s. Around that lag, the spread is only 0.075 s and 0.089 s. Exp03 could not tell which system was biased, so it applied no correction; §4.4 settles this.



- **Forced alignment.** torchaudio's MMS_FA force-aligns all 430 written words on the 16 kHz vocal stem. Emissions are computed in 30 s windows with 2 s of context per side (cropped), then one CTC Viterbi pass covers the whole song.
- **Whisper.** Re-run as in exp03. It reproduced exp03's medians exactly (0.2335 s and 0.1646 s).
- **Primary metric.** δ = median(Suno − MMS) over the 405 lead words of take 1. The 95 % CI comes from resampling the 56 lines (10 000 resamples).
- **Hypotheses.** H_S (Suno late) predicts $\delta \approx +0.23$ s. H_W (Whisper early) predicts $\delta \approx 0$ and is falsified if the CI lies above +0.05 s.
- **Anchor check.** The MMS timing itself is checked against *anchors*: acoustic onsets of line starts, hand-labelled blind on plots of the vocal stem. The plots show no candidate positions, and the window centre is jittered.
- **Decision rule.** Shift every word by $-\delta$ iff all four hold: the CI excludes 0; $|\delta| \geq$ one frame; MMS is within 0.05 s of the anchors (≥ 10 anchors); and the anchors agree on the sign.

Amendment. The pre-registered anchor rule required a ≥ 0.8 s rest before the line in both Suno's and MMS's timings, and only 1 of 56 line starts qualified. Suno's word spans are nearly contiguous (median gap 0.06 s), so the rule could not work. Before any comparison was computed, it was replaced by "all 56 lead line starts, ambiguous ones excluded by the annotator". Of the 56, 34 were labelled and 22 were marked ambiguous.

	Take 1 (video)	Take 2
δ = median(Suno − MMS) [line-cluster CI]	**+0.137 s [0.125, 0.147]**	+0.061 s [0.052, 0.070]
Whisper − MMS	−0.080 s [−0.100, −0.080]	−0.080 s [−0.100, −0.080]
Share with Suno later / spread around δ	96.0 % / 0.046 s	85.2 % / 0.044 s
Drift, Spearman ρ(time, d)	−0.002 (p = 0.96)	−0.083 (p = 0.095)
|Suno − MMS| vs random-shift control	0.140 s vs 0.563 s	0.065 s vs 0.543 s
Anchors (n = 34): Suno / MMS / Whisper − anchor	+0.220 / +0.041 / −0.292 s	–

Decision. All four conditions held: CI [0.125, 0.147]; $|\delta| = 0.137$ s; $\beta = 0.041$ s with 34 anchors; $\alpha_S = +0.220$ s [0.148, 0.347]. Every word of the video was therefore shifted by **−0.137 s** (`VIDEO_WORD_OFFSET_S`), and the video was re-rendered. Under the anchor rule as originally written, D3 would have failed and nothing would have changed.

Hypotheses. H_W is falsified. H_S survives, but with a smaller effect than it predicted: the 0.23 s Suno–Whisper lag splits into about 0.14 s of provider lateness and about 0.08 s of Whisper earliness. Whisper is earliest at line starts (−0.29 s against the anchors), which fits its known habit of absorbing a preceding pause into the next word.

Post hoc. The anchors were not used to estimate δ, so they test the correction out of sample.

- The median |Suno − anchor| fell from 0.233 s to 0.131 s; 32/34 anchors improved; one-sided Wilcoxon $p = 9 \times 10^{-7}$.
- At line starts the corrected onsets remain late by +0.083 s [0.011, 0.210].
- The lag is take-specific (0.137 vs 0.061 s), while Whisper's bias is identical on both takes. The offset is therefore a property of the provider's per-take alignment, and it is calibrated for take 1 only.

4.5 Render and verification (exp04)

Iteration before the full render. Section-midpoint stills, inspected visually, exposed six layout defects (Table in exp04), all fixed before the full render. Examples: overflowing kinetic words, a lattice-trail oscillation caused by too large a step size, and the ladder colliding with the lyrics.

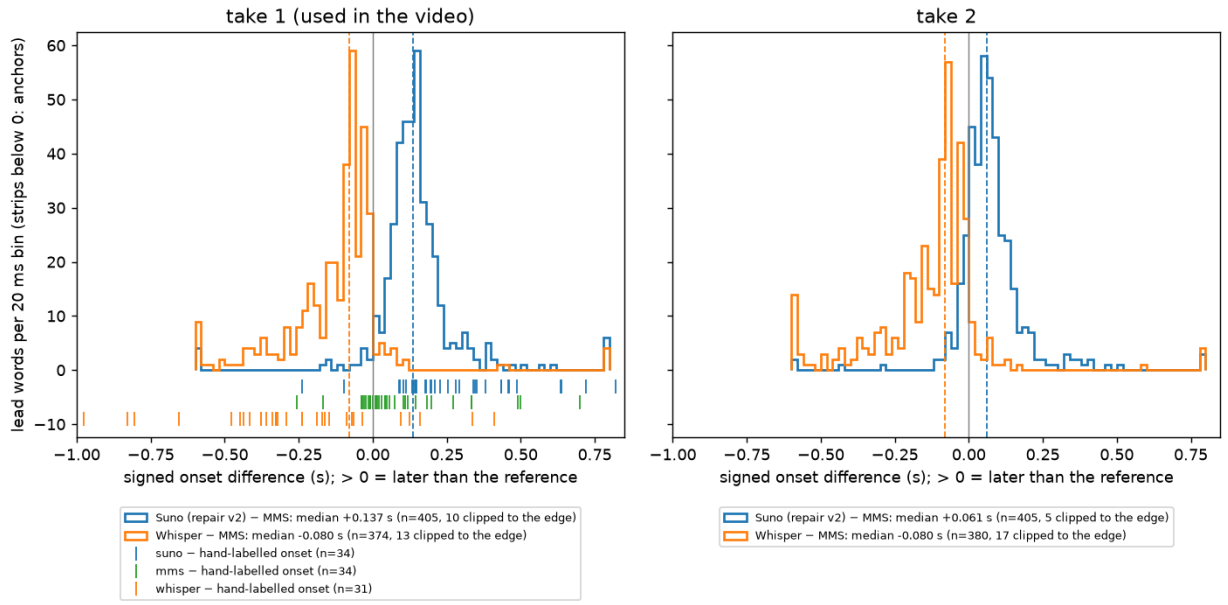


Figure 3: Exp05. Signed onset differences against the MMS forced aligner on the vocal stem (lead words) for both takes, with the hand-labelled anchors of take 1 as strips below zero.

Final render.

- 7213 frames, H.264 + AAC, 1920×1080, 30/1 fps.
- Container duration 240.45 s against a 240.42 s song, which passed the ≤ 0.5 s criterion.
- The v1.0.0 render took 347.3 s (217.6 MB) on a 16-thread desktop with 8 headless-Chrome tabs; an earlier render with v1 timings took 453.6 s.
- The v1.1.0 render with the exp05 offset took 485.0 s (217.8 MB), had the same container duration, and passed the same checks.
- The v1.2.0 render of the edited take (exp06) with the −0.156 s offset has 7212 frames and a container duration of 240.405 s for a 240.41 s song. It took 467.8 s (218.0 MB) and passed the same checks. Its audio track is 23.0 ms later than the analysed MP3 decoding.
- The MP4's audio track is 11.0 ms later than the analysed MP3 decoding (cross-correlation, first 60 s), a third of a frame. It is reported, not corrected.
- The automated check passed.
- The v1.3.0 render with the illustration track has 7212 frames and a 240.405 s container duration. It took 441.3 s (244.4 MB) and passed the same checks. Before it, a still of every card (86) was inspected. That found ten layout defects, among them overlapping graph labels, a truncated bar axis and a schematic leaving its axis (exp04, iteration 3).
- The v1.4.0 render with hero shots (render 10) has 7212 frames and a 240.405 s container duration. It took 694.1 s (263.6 MB) and passed the same checks.
  - Stills of every card at its hero moment and of every section before its cut found seven layout defects, for example labels under the hero or above the frame top. All were fixed before rendering.
  - A code review found a defect that no still can show: at hero hand-overs the camera-push origin jumped between two objects, moving the scene by up to about 70 px in one frame. It was fixed before render 10.

4.6 One-line lyric edit by section replacement (exp06)

Motivation. The author asked that the word "mind" not appear in the lyrics. It occurred once, in Verse 1 line 5. The new line is "And out of it a **thing** that learns to answer why", which keeps the meter and the rhyme. Regener-



Figure 4: Exp04. One frame from the middle of each section of the final video.

ating the song would have voided exp02–exp05 for the video's take. The provider instead offers a *replace-section* operation, which re-sings a window of ≥ 10 s of an existing take with new lyrics and, per its documentation, blends it with the rest (sunoapi.org, n.d.).

Design (pre-registered).

- **Window.** Lines 4–6, from gap midpoint to gap midpoint in audio time: 25.97–37.60 s.
- **Request.** One paid call, with segment lyrics and the full new lyrics.
- **Checks.** (a) full-length take; (b) the kept audio is the same and on the same timebase (FFT cross-correlation lag ≤ 1 ms, r ≥ 0.95); (c) the provider's word times outside the window are unchanged (median |Δ| ≤ 20 ms); (d) Whisper does not hear "mind" in the window; (e) coverage ≥ 0.99.
- **Decision.** The exp05 offset carries over only if b and c pass. Otherwise exp05 is re-run on the new take.
- **Guard.** A `banned_words` list now stops the pipeline before any paid call if the lyrics contain the word.

Amendments (both stated in the report before the analysis they govern).

1. The call returned two takes: a 240.40 s edit and a 469.52 s take that sings the song about twice. Check a was applied per take and the doubled take was dropped. The pre-specified take rule would have chosen the same take: coverage tied at 1.0, and hootCer was 0.148 vs 0.871.
2. Check b failed, so exp05's phases were re-run on the new take: Demucs + MMS_FA δ with the line-cluster bootstrap, and exp05's decision rule unchanged. The anchors were exp05's blind hand labels moved by the measured lag. An anchor was kept only where the surrounding audio is the parent's: lag-aligned per-second r ≥ 0.9 in its second and both neighbours, local lag within 1 ms of the global lag, and local r ≥ 0.9.

Results.

Check	Result
a	240.40 s (the 469.52 s take dropped)
b	**fail**: lag −17.0 ms in both kept regions; r = 0.637 (0–25 s), 0.962 (38.6–239 s); lag-aligned per-second r < 0.9 exactly in seconds 16–46
c	median |Δ| = 0.000 s (n = 401): the word times are the parent's

Check	Result
d	Whisper: "… and out of it a **thing** that learns to answer why …" (parent, same window: "… are **mine** and learns …")
e	coverage 1.0
δ re-estimated (405 lead words)	+0.156 s [0.144, 0.167], no drift (ρ = −0.02); parent +0.137 s [0.125, 0.147]
anchors	26 of 34 transferred (all 8 failures in 16–46 s); β = +0.035 s [−0.011, 0.075]; α_S = +0.221 s [0.162, 0.377]
decision	D1–D4 hold; offset **−0.156 s**

Interpretation.

- The edit is a successful causal intervention on one line, but not a local one: about 30 s of audio was regenerated.
- In the blend zone the vocal phrasing survived. On every lead line from 2 to 11, the MMS onsets moved by 0.01–0.02 s (one MMS frame) against the parent, and Whisper recovers the written words.
- The provider's word times were not re-measured outside the window. Since the file moved 17 ms earlier, Suno − MMS grew by the same amount.
- On the same 26 anchors, α_S grew by 17 ms (+0.204 → +0.221 s), while β, which compares two audio-derived onsets, stayed within one frame (+0.041 → +0.035 s). This is what a pure file shift predicts.

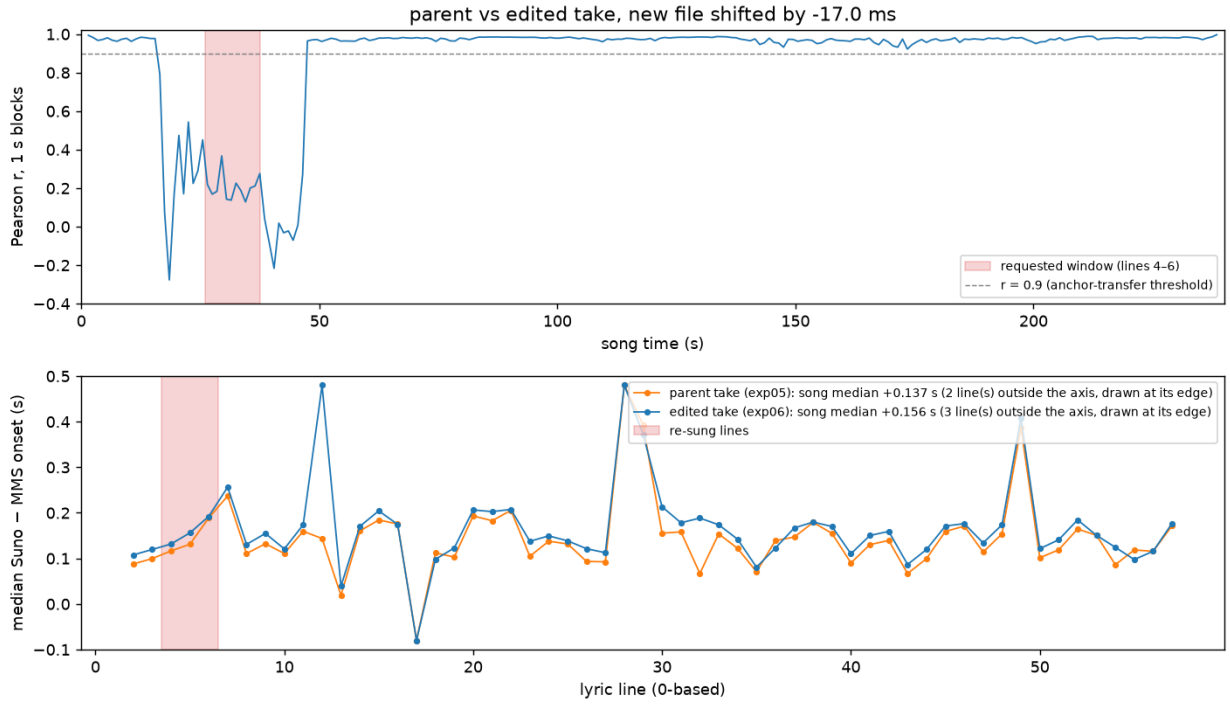


Figure 5: Exp06. Top: lag-aligned per-second waveform correlation between the parent and the edited take; the shaded band is the requested window, and the provider changed audio from about 16 s to 46 s. Bottom: per-line median Suno − MMS onset on both takes.

4.7 Figure cards from released data (exp07)

Pre-specified checks (fixed in the plan before the data phase ran): 1. The five largest Fourier frequencies of the released epoch-40k embedding equal the paper's key frequencies. 2. ARENA's five highest induction scores include IOI's induction heads 5.5 and 6.9 (Wang et al., 2022). 3. Every on-screen number matches its source. 4. Coverage is 1.0. 5. Paired frames of v1.3.0 compress to more PNG bytes than v1.2.0 frames at the same timestamps: median $\log_2$ ratio > 0, with a section-bootstrap 95 % CI excluding 0.

The fifth check uses a standard proxy for visual complexity (Machado et al., 2015).

Check	Result
1	**pass**: {14, 35, 41, 42, 52}; the 6th largest is 31
2	**pass**: 5.5 = 0.909, 6.9 = 0.898, 5.1 = 0.897, 7.10 = 0.891, 7.2 = 0.825
3	**pass after corrections**: a Golden Gate quote and clamp value, an ActAdd layer and coefficient, and a "2–6 = well trained" band for WeightWatcher's α were in our plan but not in the cited sources, and were removed or corrected
4	**pass**: 56/56 lines, 86 cards, the shortest on screen for 2.0 s
5	**pass**: +0.263 [+0.177, +0.422]; frame bootstrap [+0.208, +0.343]; 114/120 frames larger; per section from +0.003 (verse 1) to +0.634 (bridge)

Descriptive numbers from the released data (not pre-registered): - **Weight decay.** With weight decay 1.0, test loss first drops below 1 at epoch 9,890; with 0.1, at epoch 90,890. The runs also differ in seed, and the 0.1 run's stored config is stale. - **Double descent.** Test error falls to 0.219 at width 4, peaks at 0.303 at width 10, and reaches 0.163 at width 64 (5 trials). - **Refusal.** Ablating the refusal direction raises JailbreakBench answers without a refusal phrase from 5 % to 100 % (LlamaGuard-2 judge: 3 % to 82 %).

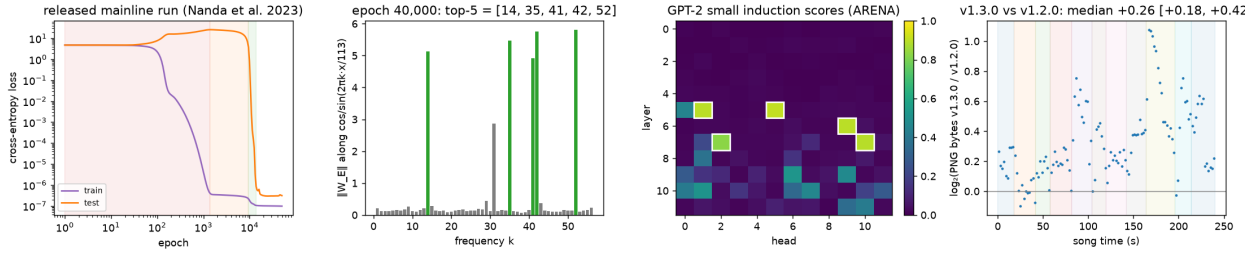


Figure 6: Exp07. (a) Train and test loss of the released grokking run with the paper's three phases. (b) Fourier norms of its epoch-40k embedding; green bars are the paper's key frequencies. (c) ARENA's induction scores for GPT-2 small; ringed are the top five. (d) Per-frame $\log_2$ PNG-size ratio v1.3.0 / v1.2.0 with sections shaded.

4.8 Figure cards inside the scenes (exp08)

Pre-specified checks (committed before the v1.4.0 render, `reports/exp08_integration.md`): 1. The render verifies. 2. Non-inferiority of detail against the released v1.3.0 MP4, whose sha256 equals GitHub's asset digest. The test passes if the section-bootstrap 95 % CI lower bound of the median paired $\log_2$ PNG-size ratio is above −0.10. A margin fixed in advance and compared with the CI bound is the CONSORT form (Piaggio et al., 2012). 3. Design checks computed from the timeline with a Python mirror of the lifecycle: - hero area ≤ 0.6 of the frame; - hero time ≤ 3.2 s plus half an ease; - at most 4 cards active.

The mirror reads its constants from the TypeScript source, and it agreed with it exactly on 2,400 samples.

Check	Result
1	**pass** (render 10, above)
2	**pass**: −0.014 [−0.075, +0.016]; frame bootstrap [−0.043, −0.002]; 47/120 frames larger; per section from −0.191 (verse 1) to +0.133 (finale)
3	**pass**: hero area 0.328 (0.193 with giant lyrics), longest hero 3.20 s, at most 4 active (6 in motion counting cards flying back to dock)

Descriptive. - **Hero time.** A figure is the hero (heroness ≥ 0.5) during 82.6 % of sung time and 74.6 % of the song. The scene has the frame alone for 25.4 % of the song. - **Coverage.** Mean nominal hero coverage over sung time is 23.3 % of the frame. - **Docking.** 71 of the 86 cards dock into a scene object; the 15 cards of the three fly-through scenes fly past the camera.

Interpretation, post hoc and untested. In verse 1, the bridge and verse 2, v1.3.0 showed up to four full-brightness corner cards at once. v1.4.0 shows one hero over a scene dimmed to 72 % brightness. Dimming lowers PNG size without removing any element, so the proxy confounds detail with brightness there. An ablation with the dimming off would separate the two, and it was not run.

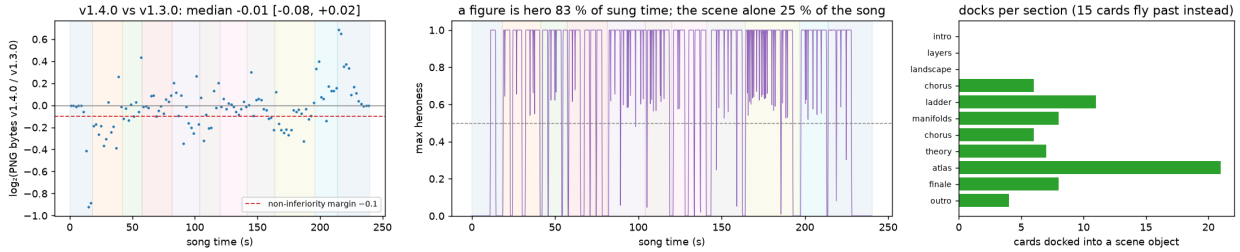


Figure 7: Exp08. (a) Per-frame $\log_2$ PNG-size ratio v1.4.0 / v1.3.0 with sections shaded and the pre-specified margin. (b) Largest heroness over the song with the 0.5 threshold. (c) Cards docked into a scene object per section.

4.9 Figures as part of their scenes (exp09)

Pre-specified (`reports/exp09_unified.md`, committed before the render): - **H0:** the render verifies. - **H1:** every panel is shown ≥ 3 s, or ≥ 2.5 s when its trigger is within 3 s of a cut. - **H2:** at most two panels at once. - **H3:** no slot overlaps a protected region or band. - **H4:** at 13–18 s, the median $\log_2$ PNG ratio against v1.3.0 is ≥ 0. - **H5:** non-inferiority of detail against v1.4.0 before the outro, with margin 0.10.

Check	Render 11 (pre-registered)
H0	**pass**
H1	**pass**: shortest panel 2.76 s (cut-limited), median 5.8 s
H2	**pass**: 2
H3	**pass**: no overlap
H4	**fail**: −0.58 (six frames, −0.55 to −0.59)
H5	**pass**: +0.027 [−0.049, +0.088]; verse 1 +0.21, bridge −0.22

Post-hoc (render 12, released). After H4 failed, the planned match cut was added: verse 1's layers fly out behind the tokens card. H4 is now −0.093 (−0.156 to −0.061), still below the pre-registered ≥ 0, so it is reported as not met. H5 is +0.033 [−0.047, +0.102]. Against v1.4.0, the window gained +0.26, +0.83 and +0.80 $\log_2$ at 13, 15 and 17 s.

Descriptive. The median time a figure is shown rose from 2.22 s (v1.4.0 hero time) to 6.0 s (v1.5.0, on screen at full panel size). Figures under 2 s fell from 41/86 to 1/82: the closing banner, 0.94 s. The share of figure time over a protected object fell from 100 % to 0 % by construction.

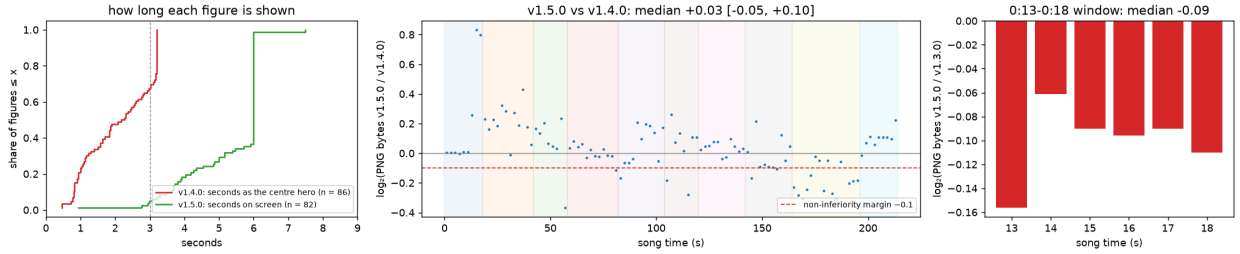


Figure 8: Exp09. (a) Empirical CDF of the seconds each figure is the hero (v1.4.0) or on screen (v1.5.0). (b) Per-frame $\log_2$ PNG-size ratio v1.5.0 / v1.4.0 before the outro, with the margin. (c) The 0:13–0:18 window against v1.3.0.

5. Discussion

What the evidence supports.

- The pipeline renders a valid, song-length video deterministically.
- Its beat times are accurate to under one frame on click tracks between 70 and 160 BPM.
- Its word onsets agree with an independent recogniser to about a quarter of a second (median), clearly better than chance. Most of that disagreement is a constant 0.17–0.23 s lag, around which relative timing agrees to about 0.08 s (post-hoc).
- That lag is shared between the two systems. Against a lyrics-aware forced alignment of the separated voice, the provider is 0.137 s late on the original video take and Whisper is 0.080 s early. A pre-registered global correction of −0.137 s moved 94 % of independently hand-labelled line starts closer.
- Concrete content can be added without re-running anyone's experiment. Released artifacts covered every "real data" card, and the two checks that could fail against the papers passed.
- Concrete examples can be staged inside the general animation without losing measurable detail overall (exp08). The change redistributes detail between sections rather than adding it.
- A per-scene layout contract (slots, protected objects, lyric bands, tested) satisfies the layout properties meant to address the author's "two overlapped videos" judgement, and keeps detail (exp09). Whether viewers perceive the video as more unified was not measured.
- The video's line was changed with a one-call section replacement. Because the provider shifted the file and copied its old word times, the correction had to be re-estimated (−0.156 s) and could not be carried over.

What it does not support.

- Any claim about musical quality.
- Any general claim about what replace-section changes: one call is an existence proof, not a characterisation.
- Any general claim about Suno's alignment accuracy. The lag differed between the two takes of one generation (0.137 vs 0.061 s), so a per-take measurement is needed, not a constant.
- Frame-exact karaoke. The forced aligner itself sits about 0.04 s after the hand-labelled onsets, and line starts remain about 0.08 s late after the correction.
- The beat accuracy on the actual orchestral mix, which has no ground truth.
- Any claim that the denser video is better or easier to follow: compressed size measures detail, not quality, and no viewer study was run. The same holds for the claim that hero shots integrate the examples better than corner boxes: exp08 measures the staging, not its effect on viewers.

The most useful outcome was negative. A reasonable-looking heuristic, written to fix visible artifacts, made

things worse on 43 words. Only an independent reference and a pre-stated test revealed this. The general lesson for code-animated music videos is to treat every timing transformation as an intervention and to measure it against something independent, not against the same signal it was designed from.

6. Limitations

- **Sample size.** One generation (n = 1 task, 2 takes) and one song. Nothing here estimates population-level behaviour of Suno or of the provider.
- **Reference quality.** The Whisper reference is noisy: sung vocals over an orchestral mix, and timestamps from cross-attention. Exp05 therefore added a lyrics-aware forced aligner on a separated vocal stem.
  - That aligner was trained on read speech. Its absolute timing was checked only at 34 line starts.
  - The anchors were hand-labelled by the agent from plots of the vocal stem, not by listening, and 22 of 56 were too ambiguous to label.
  - The anchor-selection rule had to be amended after its first run, though before any comparison.
- **Synthetic beat test.** Click tracks lack the onsets and tempo drift of real music, and downbeats rely on a 4/4 heuristic that was not validated.
- **Exp06 amendments.** Both were written after seeing part of the data: the duration check after seeing the two durations, and the anchor transfer after seeing check b. The transfer reuses labels made on the parent take. It is valid only where the audio is the parent's shifted by a constant, which is checked per anchor. No new blind labels were made for the edited take.
- **Post-hoc analyses.** Repair v2 and the slope tempo estimator were designed after seeing data. Both are labelled as post-hoc throughout, and no confirmatory claim is made about them.
- **Illustration track.** Several released files state no licence. Only small derived numbers are committed, with attribution on screen.
  - Schematic cards illustrate ideas; they are not data.
  - The weight-decay comparison rests on one run per setting, with different seeds.
  - Legibility of 2–6 s cards was judged by the agent from stills, not by viewers.
- **Hero shots (exp08).**
  - The hero area is a nominal box, not measured pixels.
  - The density proxy confounds detail with brightness where the scene is dimmed.
  - The hero share (82.6 % of sung time) is a design choice for a maximalist video. It was measured but not tuned against any viewer outcome.
- **Scene-owned figures (exp09).**
  - The protected regions are static rectangles, so moving 3D objects can still pass under a panel.
  - The intro fix was post hoc, made after the pre-registered H4 failed.
  - The rewind's outro is excluded from the density test.
  - The curation removed four cards and moved 32 into their scenes; this is a judgement, logged with reasons.
- **Black-box components.** Schematic content is labelled per card. The music model and the provider are black boxes, and hootCer's definition is undocumented.

7. Conclusion

A code-animated music video can be built as an ordinary, testable data pipeline:

- the paid generative step is cached;
- the timing components are validated against references fixed in advance;
- the render is deterministic and checked automatically.

Doing so turned up a quantization bug, a half-tempo failure mode, a harmful repair heuristic, and a 0.137 s lag in the provider's word onsets that is now corrected. A later one-line lyric edit added a fifth finding: the provider's edit moved the whole file by 17 ms and regenerated far more than the requested window, while reporting the old word times. None of these would have been visible from the finished video's description. Adding concrete,

cited content showed the same pattern: a fact-check of our own plan against the cited sources removed three overclaims before release. Moving that content into the scenes did too: a code review found a camera jump that no still frame could show, and a pre-registered density check surfaced a section that lost detail, which the aggregate hides. Making each scene own its figures then fixed the cover and dwell problems by construction, and a failed pre-registered check pointed at a planned element that had been skipped. Code, data, logs-derived reports and the video are available at https://github.com/BurnyCoder/ambitious-science-of-information-processing-anthem.

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Appendix A. Reproduction

```
uv sync && (cd video && npm install)
cp .env.example .env          # key only needed for a new generation
uv run anthem                 # cached song -> timeline -> render -> verify
uv run python scripts/validate_beats.py
uv run --group analysis python scripts/validate_alignment.py small.en
uv run --group analysis python scripts/validate_onset_offset.py  # exp05; reuses the committed labels
uv run --group analysis python scripts/validate_lyric_edit.py    # exp06; uses the committed data/song/ of b
uv run --group analysis python -I scripts/vignette_data.py     # exp07 data phase (downloads 31 pinned files
uv run python -I scripts/validate_density.py              # exp07 density (needs the v1.2.0 MP4 in out/v1.2.
uv run python -I scripts/validate_density.py out/v1.3.0/we-will-know-anthem.mp4 out/anthem.mp4 -
-name exp08_density --margin 0.10
uv run python -I scripts/validate_unified.py                                  #  exp09  H1-
H4 (exp08's integration script: git checkout v1.4.0)
uv run python -I scripts/validate_density.py out/v1.4.0/we-will-know-anthem.mp4 out/anthem.mp4 -
-name exp09_density --margin 0.10 --until 213.96
uv run python -I scripts/make_vignette_docs.py             # docs/vignettes.md, the card catalog
uv run --group analysis python scripts/make_figures.py
```

The full list of the 86 cards with times, triggers, badges and sources is in `docs/vignettes.md`.

Appendix B. Lyrics structure

The lyrics have 11 sections: Intro, Verse 1, Pre-Chorus, Chorus, Verse 2, Pre-Chorus, Chorus, Verse 3, Bridge, Final Chorus, Outro. The full text is in `content/lyrics.txt`, and the section-to-scene map is in `content/scenes.toml`, described in `docs/scenes.md`.

AI statement

Claude Code (Anthropic) wrote this codebase and this paper, executing an exact specification provided by a human (Libor Burian). Human cleanup and rewriting are in progress.